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324-6

Call the form f (left) non-degenerate if 0 is the only vector α such that $f(\alpha, \beta) = 0 \forall \beta$. Let f be a form on V , prove that f is non-degenerate iff T_f is non-singular.

Proof: We have $f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle$, so $L_f: V \rightarrow V^*, \alpha \mapsto \langle T_f(\alpha), \cdot \rangle$. Hence $\text{Ker} L_f = \text{Ker} T_f(\alpha)$, and f is non-degenerate iff T_f is non-singular.

324-7

Let f be a form on $V, \dim V < \infty$. Define right non-degeneracy and prove that f is left non-degenerate iff f is right non-degenerate.

Proof: Define f is right non-degenerate iff 0 is the only β such that $f(\alpha, \beta) = 0 \forall \alpha$. Hence f is right non-degenerate iff $0 = (\text{Im} T_f)^\perp$, which is equivalent to $\text{Im} T_f = V$ or T_f is surjective. Since T_f is surjective iff T_f is non-singular, f is left non-degenerate iff f is right non-degenerate.

324-8

Let f be a non-degenerate form on $\dim V < \infty, L \in V^*$. Show that there exists a unique β such that $L(\alpha) = f(\alpha, \beta)$ for all α .

Proof: f is non-degenerate so there is an isomorphism T_f such that $f(\alpha, \beta) = \langle T_f(\alpha), \beta \rangle = \langle \alpha, T_f^* \beta \rangle$. By Riesz representation theorem, there is a unique γ such that $L(\alpha) = \langle \alpha, \gamma \rangle$, so $\beta = (T_f^*)^{-1}(\gamma)$ is unique.

324-9

Let f be a non-degenerate form on $\dim V < \infty$. Show that each linear operator S has an adjoint relative to f , i.e., S' such that $f(S\alpha, \beta) = f(\alpha, S'\beta) \forall \alpha, \beta$.

Proof: It suffices to find S' such that

$$\langle T_f(S\alpha), \beta \rangle = \langle T_f(\alpha), S'\beta \rangle,$$

i.e., $\langle \alpha, S^* T_f^* \beta \rangle = \langle \alpha, T_f^* S' \beta \rangle$. Hence we can take $S' = (T_f^*)^{-1} S^* T_f^* = (T_f^{-1} S T_f)^*$.

330-1

Let $V = \mathbb{C}^2$, for which $\alpha \in V$ is there a positive linear operator T such that $\alpha = T e_1$?

Solution: Since T is positive, $\langle \alpha, e_1 \rangle = \langle T e_1, e_1 \rangle \geq 0$. Suppose $\alpha = a e_1 + b e_2$, then $T = \begin{pmatrix} a & \bar{b} \\ b & c \end{pmatrix}$, hence by positivity, $a, c \geq 0$ and $ac \geq |b|^2$. Hence $a > 0$ or $\alpha = 0$.

If $a > 0$, then choose $T = \begin{pmatrix} a & \bar{b} \\ b & |b|^2/a \end{pmatrix}$ we have $\alpha = T e_1$. Hence α satisfy $\langle \alpha, e_1 \rangle > 0$ or $\alpha = 0$.

330-2

Let $V = \mathbb{R}^2$ and $\theta \in \mathbb{R}$, $T_\theta(x_1, x_2) = (x_1 \cos \theta - x_2 \sin \theta, x_1 \sin \theta + x_2 \cos \theta)$. For which values θ is T_θ positive?

Solution: $T_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$, so T_θ is positive iff $\cos \theta > 0$.

330-9

Prove that every entry on the main diagonal of a positive matrix is positive.

Proof: For a positive matrix A , $A_{ii} = \langle Ae_i, e_i \rangle > 0$.

330-11

Let $\dim V < \infty$ and E the orthogonal projection of $V \rightarrow W$.

(a) Prove that for any $c > 0$, $cI + E$ is positive.

(b) Express in terms of E a self-adjoint linear operator T such that $T^2 = I + E$.

Proof: (a) For any $v \in V$, $\langle (cI + E)v, v \rangle = \langle Ev + cv, v \rangle$. Suppose $v = u + w \neq 0$ where $w \in W$ and $u \in W^\perp$, then $\langle Ev + cv, v \rangle = \langle (c + 1)w + cu, w + u \rangle = (c + 1)\langle w, w \rangle + c\langle u, u \rangle > 0$.

(b) Since $E^2 = E$, take $T = I + (\sqrt{2} - 1)E$, then $T^2 = I + 2(\sqrt{2} - 1)E + (\sqrt{2} - 1)^2E = I + E$. Clearly T is self-adjoint since both I and E are self-adjoint.

330-13

Let A be a self-adjoint $n \times n$ matrix, prove that there is a real number c such that $cI + A$ is positive.

Proof: Let c be greater than the absolute value of all eigenvalues of A , then for any eigenvalue $(cI + A)v = \lambda v$, $Av = (\lambda - c)v$ so $\lambda > 0$. Hence $cI + A$ is positive.

330-14

Prove that the product of two positive linear operators is positive iff they commute.

Proof: If they commute, they can be simultaneously orthogonally diagonalized, hence their product is still positive.

Conversely, if TS is positive, it is self-adjoint, so $TS = (TS)^* = S^*T^* = ST$, hence they commute.

330-15

Let S, T be positive operators, prove that every eigenvalue of ST is positive.

Proof: Note that $T^{1/2}(ST)T^{-1/2} = T^{1/2}ST^{1/2}$ where $T^{1/2}$ is invertible, so ST and $T^{1/2}ST^{1/2}$ have the same eigenvalues. Note that $T^{1/2}ST^{1/2} = (T^{1/2}S^{1/2})(S^{1/2}T^{1/2})$ is positive, so every eigenvalue is positive.

347-4

Let $T \in \mathcal{L}(V)$ prove that TFAE:

- (a) T is normal; (b) $|T\alpha| = |T^*\alpha| \forall \alpha \in V$; (c) $T = T_1 + iT_2$ where T_1, T_2 are self-adjoint and $T_1T_2 = T_2T_1$;
- (d) If $\alpha \in V, c \in F$ such that $T\alpha = c\alpha$ then $T^*\alpha = \bar{c}\alpha$;
- (e) There is an orthonormal basis consisting of eigenvectors of T ;
- (f) There is an orthonormal basis such that $[T]_{\mathcal{B}}$ is diagonal;
- (g) There is a polynomial $g \in \mathbb{C}[x]$ such that $T^* = g(T)$;
- (h) Every T -invariant subspace is also T^* -invariant;
- (i) $T = NU$ where N is non-negative, U is unitary, and $UN = NU$;
- (j) $T = c_1E_1 + \dots + c_kE_k$ where $I = E_1 + \dots + E_k, E_iE_j = 0 \forall i \neq j$ and $E_i^2 = E_i = E_i^*$.

Proof: (a) $\Leftrightarrow$ (b)

$$|T\alpha|^2 - |T^*\alpha|^2 = \langle \alpha, (T^*T - TT^*)\alpha \rangle.$$

Hence $|T\alpha| = |T^*\alpha| \forall \alpha \iff T^*T = TT^*$.

(a) $\Leftrightarrow$ (c) If $T = T_1 + iT_2$ where T_1, T_2 self-adjoint and commutative, $T^* = T_1 - iT_2$ so $TT^* = T_1^2 + T_2^2 = T^*T$ hence T is normal.

If T is normal, then $T_1 = \frac{T+T^*}{2}$ and $T_2 = \frac{T-T^*}{2i}$ commute and $T = T_1 + iT_2$.

(a) $\Rightarrow$ (d) If T is normal, and $T\alpha = c\alpha$, then $T - cI$ is normal so $|(T - cI)\alpha| = |(T - cI)^*\alpha|$. Hence $T\alpha = c\alpha$ implies $T^*\alpha = \bar{c}\alpha$.

(d) $\Rightarrow$ (e) If $T\alpha = c\alpha \implies T^*\alpha = \bar{c}\alpha$, then take any eigenvector e_1 of $T, \langle e_1 \rangle^\perp$ is a T -invariant subspace and also T^* -invariant. Using induction, we know that $T|_{\langle e_1 \rangle^\perp}$ has a orthonormal basis consisting of eigenvectors, hence T has a orthonormal basis consisting of eigenvectors.

(e) $\Rightarrow$ (f) Trivial. (f) $\Rightarrow$ (a) Trivial.

(f) $\Rightarrow$ (g) Trivial by Lagrange interpolation. (g) $\Rightarrow$ (h) Trivial. (h) $\Rightarrow$ (d) If $T\alpha = c\alpha$ then $\langle \alpha \rangle$ is T -invariant, hence T^* -invariant. Suppose $T^*\alpha = \lambda\alpha$, then $\langle T\alpha, \alpha \rangle = \langle \alpha, T^*\alpha \rangle$ implies $\lambda = \bar{c}$.

(f) $\Rightarrow$ (i) Take an orthonormal bases e_i such that $Te_i = c_i e_i$. Let $Ne_j = |c_j|e_j$ and $Ue_j = \text{sgn}(c_j)e_j$, then $T = UN$ and $UN = NU$.

(i) $\Rightarrow$ (a) Suppose $T = NU = UN$, then $T^* = U^*N = NU^*$. Hence $TT^* = N^2 = T^*T$.

(j) $\Rightarrow$ (f) Take orthonormal basis of $E_1, \dots, E_k$, combining them gives an orthonormal basis such that $[T]_{\mathcal{B}}$ is diagonal as $\text{diag}(c_1, \dots, c_1, \dots, c_k)$.

(f) $\Rightarrow$ (j) If $[T]_{\mathcal{B}}$ is diagonal, take all eigenvalues $c_1, \dots, c_k, W_j = \text{Ker}(T - c_jI)$ and E_j the orthogonal projections to W_j . Then $T = c_1E_1 + \dots + c_kE_k$

347-6

Let V be a complex inner product space and U a unitary operator such that $U\alpha = \alpha \implies \alpha = 0$. Let $f(z) = i\frac{1+z}{1-z}$, and show that

- (a) $f(U) = i(I + U)(I - U)^{-1}$; (b) $f(U)$ is self-adjoint; (c) for every self-adjoint operator $T \in \mathcal{L}(V), U = (T - iI)(T + iI)^{-1}$ is unitary and $T = f(U)$.

Proof: (a) $U\alpha = \alpha \implies \alpha = 0$ means $U - I$ is invertible, so $f(U) = i(I + U)(I - U)^{-1}$ is well-defined.

(b) $f(U)^* = -i(I + U^*)(I - U^*)^{-1} = -i(I + U^{-1})(I - U^{-1})^{-1} = f(U)$ so $f(U)$ is self-adjoint.

(c) For any self-adjoint $T, U^* = (T + iI)(T - iI)^{-1} = U^{-1}$ is unitary, and $f(U) = i(I + U)(I - U)^{-1} = T$.

331-5

Let $A = \begin{pmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{3} \end{pmatrix}$.

(a) Show that A is positive.

(b) Let $V = \mathbb{R}^{2 \times 1}$, with inner product $\langle X, Y \rangle = Y^t A X$. Find an orthonormal basis for V , by applying Gram-Schmidt process to e_1, e_2 .

(c) Find an invertible 2×2 real matrix P such that $A = P^t P$.

Proof: (a) Clearly A is self-adjoint, $A_{11} > 0$ and $\det A = \frac{1}{12} > 0$. Hence A is positive.

(b) $\langle e_1, e_1 \rangle = e_1^t A e_1 = 1$ so $\alpha_1 = e_1$. $\langle e_1, e_2 \rangle = \frac{1}{2}$ so $\beta_2 = e_2 - \langle e_1, e_2 \rangle e_1 = (-\frac{1}{2}, \frac{1}{2})$, and $\alpha_2 = \frac{1}{\sqrt{3}}(-1, 1)$.

The orthonormal basis is $\left\{ (1, 0), \frac{1}{\sqrt{3}}(-1, 1) \right\}$.

(c) We need to find P such that $\langle X, Y \rangle = (PY)^t(PX)$, hence $P = \begin{pmatrix} 1 & -\frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{3}} \end{pmatrix}$ is sufficient.

331-10

Let $\dim V < \infty$, $T, U \in \mathcal{L}(V)$. We write $T < U$ if $U - T$ is a positive operator. Prove that:

(a) $T < U$ and $T > U$ is impossible;

(b) If $T < U$ and $U < S$ then $T < S$;

(c) If $T < U$ and $0 < S$, it need not be that $ST < SU$.

Proof: (a) If $T < U$ and $T > U$ then $A = U - T$ and $-A$ are both positive, so $0 = A + (-A)$ is positive, a contradiction.

(b) If $T < U$ and $U < S$, then $(S - T) = (S - U) + (U - T)$ is positive, so $T < S$.

(c) We find two positive matrix $S, P = U - T$ whose product is not positive.

Consider $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, then $AB \neq BA$ while both are positive, hence AB is not positive (not even self-adjoint).